

Applied Machine Learning — Week 7

Unsupervised Learning · Clustering (K-Means, Agglomerative, GMMs)

Comprehensive Revision Notes

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Unsupervised Learning: Overview

In unsupervised learning, we have data with **no labels**: $\mathcal{D} = \{x_1, \dots, x_N\}$, $x_i \in \mathbb{R}^D$.

Unsupervised Tasks

- **Density estimation**: model the distribution $p(x)$.
- **Clustering**: find discrete groups that summarise the data.
- **Dimensionality reduction**: find a low-dimensional subspace/manifold that captures most variation.
- **Feature learning**: discover useful representations for downstream tasks.

Evaluation is harder than supervised learning: there is no “ground truth.” Options include likelihood on held-out data (probabilistic models), comparison with known labels if available, or performance on downstream tasks.

K-Means Clustering

Objective

K-Means Objective

Partition N data points into K clusters $C_1, \dots, C_K$ to minimise the **within-cluster sum of squares (WCSS)**:

$$\text{WCSS} = \sum_{k=1}^K \sum_{x_i \in C_k} \|x_i - \mu_k\|^2, \quad \text{where } \mu_k = \frac{1}{|C_k|} \sum_{x_i \in C_k} x_i.$$

Algorithm

K-Means Algorithm (Lloyd's)

1. **Initialise**: choose K initial cluster centres $\mu_1, \dots, \mu_K$ (e.g. random data points, or K-means++).
2. **Assign**: for each x_i , assign to the nearest centre: $c(x_i) = \arg \min_k \|x_i - \mu_k\|^2$.
3. **Update**: recompute each centre: $\mu_k = \frac{1}{|C_k|} \sum_{x_i \in C_k} x_i$.
4. **Repeat** steps 2–3 until no assignments change.

Converges in finite steps (WCSS decreases or stays the same at each step). Finds a **local optimum** — run with multiple random initialisations.

Limitations

△ K-Means Limitations

- **Only spherical clusters**: cluster shapes are Voronoi diagrams of the centres — cannot capture elongated or non-convex shapes.
- **Unequal density/size**: K-means tends to split large clusters and merge small ones.
- **Sensitive to initialisation**: different initialisations $\Rightarrow$ different solutions.
- **Sensitive to scaling**: features on larger scales dominate the distance.
- **No principled way to choose K** : try different values, use elbow method or silhouette scores.

- **Centres are not data points:** the mean may not resemble any actual data point (especially for discrete/categorical data).

K-Means as Feature Extractor

After fitting K-means, create features for new points:

- **Cluster membership:** categorical feature (which cluster?).
- **Distances to centres:** K real-valued features ($\|x - \mu_1\|, \dots, \|x - \mu_K\|$).

Useful nonlinear feature space, especially for low-dimensional data.

K-Medoids Variant

Constrains centres to be actual data points (the “medoid” — the point minimising average distance to all others in the cluster). More robust to outliers; works with arbitrary dissimilarity functions; but higher computational cost.

Agglomerative (Hierarchical) Clustering

Agglomerative Clustering

Bottom-up: start with N singleton clusters. At each step, merge the two most similar clusters. Produces a **dendrogram** (tree of merges). Cut the dendrogram at a desired level to get K clusters.

Linkage Rules

Given a point-level dissimilarity $d(x, x')$, define cluster-level dissimilarity $D(G, H)$:

Linkage Rules

- **Single linkage** (nearest-neighbour): $D_{SL}(G, H) = \min_{x \in G, x' \in H} d(x, x')$.
Tends to produce “chaining” — elongated clusters. Can find non-convex shapes.
- **Complete linkage** (furthest-neighbour): $D_{CL}(G, H) = \max_{x \in G, x' \in H} d(x, x')$.
Produces compact, roughly spherical clusters.
- **Average linkage:** $D_{GA}(G, H) = \frac{1}{|G||H|} \sum_{x \in G} \sum_{x' \in H} d(x, x')$.
Compromise between single and complete.
- **Ward’s linkage:** merge clusters that produce the *smallest increase* in total within-cluster variance:

$$D_W(G, H) = \frac{1}{|G| + |H|} \sum_{x_i \in G \cup H} \|x_i - \mu_{G \cup H}\|^2 - \frac{1}{|G|} \sum_{x_i \in G} \|x_i - \mu_G\|^2 - \frac{1}{|H|} \sum_{x_i \in H} \|x_i - \mu_H\|^2.$$

Tends to give **evenly-sized** clusters. Resembles K-means conceptually.

Gaussian Mixture Models (GMMs)

The Model

Gaussian Mixture Model

$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k),$$

where $\pi_k \geq 0$, $\sum_k \pi_k = 1$ are the **mixture weights**, and each component k is a Gaussian with mean μ_k and covariance Σ_k .

Latent variable formulation:

$$z_n \sim \text{Discrete}(\pi_1, \dots, \pi_K),$$
$$x_n \mid z_n=k \sim \mathcal{N}(\mu_k, \Sigma_k).$$

z_n is a latent variable indicating the cluster assignment of point x_n .

EM Algorithm (Intuitive)

EM for GMMs

Initialise μ_k, Σ_k, π_k (e.g. run K-means first for μ_k).

E-step (soft assignment): compute the “responsibility” of each component k for each point n :

$$\gamma_{nk} = p(z_n=k \mid x_n) = \frac{\pi_k \mathcal{N}(x_n \mid \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(x_n \mid \mu_j, \Sigma_j)}.$$

M-step (update parameters): using the soft assignments as weights:

$$N_k = \sum_{n=1}^N \gamma_{nk}, \quad \mu_k = \frac{1}{N_k} \sum_n \gamma_{nk} x_n, \quad \Sigma_k = \frac{1}{N_k} \sum_n \gamma_{nk} (x_n - \mu_k)(x_n - \mu_k)^\top, \quad \pi_k = \frac{N_k}{N}.$$

Iterate until convergence. Each iteration is guaranteed to increase (or maintain) the log-likelihood.

Connections:

K-Means as a limiting case of GMM K-means is GMM with $\Sigma_k = \sigma^2 I$ for all k and $\sigma \rightarrow 0$. In this limit, the soft assignments γ_{nk} become *hard* (0 or 1), recovering the K-means assignment step. The M-step reduces to computing unweighted cluster means.

GMM vs. K-Means

	K-Means	GMM
Cluster shape	Spherical (Voronoi)	Ellipsoidal (per-component Σ_k)
Assignment	Hard	Soft (probabilistic)
Cluster sizes	Tend to be equal	Can vary (π_k)
Model	Non-probabilistic	Probabilistic (likelihood)
Choosing K	Elbow/heuristic	Likelihood on held-out data / BIC
Initialisation	Sensitive	Sensitive (use K-means to initialise!)

Thoughts on Mixture Models

Pros: natural probabilistic model; generalises to other data types (Bernoulli, Poisson, Student- t); different likelihoods per dimension; choosing K via likelihood or Bayesian approaches.

Cons: only finds “compact” clusters; non-convex objective (initialisation-dependent); GMMs sensitive to outliers (use Student- t mixtures instead).

Practical Considerations

△ Scaling and Clustering

K-means uses Euclidean distance, so features on larger scales dominate. Rescaling / normalising dimensions can help — but it can also **remove** obvious structure if the scale differences are informative. Think carefully before standardising.

Combinatorial explosion: the number of ways to assign N points to K clusters is a Stirling number $S(N, K)$. For $K = 4, N = 10$: 34,105 possible clusterings. For $N = 19$: $\sim 10^{10}$. Exhaustive search is impossible; hence heuristic algorithms.

Exam-Style Questions & Lecture Exercises

Exam Q&A:

Q1 — K-Means Centres (2 marks) **Q:** After K-means convergence, are cluster centres always original data points?

A: False. Centres are *means* of assigned points, not actual data points. Use K-medoids if you need centres to be data points.

Exam Q&A:

Q2 — Clustering Algorithm Identification (Practice Exam Q2) **Q:** For each clustering pattern, identify which among K-means, single-linkage, and GMM could produce it.

A: (A) Two well-separated compact spherical clusters: K-means, single linkage, GMM (all three). **(B) Outlier assigned to the closer of two clusters:** K-means (always assigns to nearest centre); GMM if the outlier is within range. Single linkage could chain it. **(C) Elongated/non-spherical cluster split:** Only single linkage (chains); K-means and spherical GMM would split it. **(D) Clusters of different sizes:** GMM (different Σ_k, π_k); K-means struggles.

Exam Q&A:

Q3 — GMM vs K-Means (3 marks) **Q:** Give three advantages of GMMs over K-means.

A: (1) Soft (probabilistic) assignments quantify uncertainty about cluster membership. (2) Per-component covariance Σ_k allows ellipsoidal clusters, not just spheres. (3) Mixture weights π_k allow different cluster sizes. Additionally, the likelihood provides a principled way to compare different values of K .

Exam Q&A:

Q4 — Linkage Rules (3 marks) **Q:** Compare single and complete linkage. Which produces more compact clusters?

A: Single linkage uses the *minimum* pairwise distance, which can cause “chaining” (merging clusters connected by a single close pair of points), producing elongated clusters. Complete linkage uses the *maximum* pairwise distance, so it only merges clusters where *all* pairs are close, producing compact, roughly spherical clusters. Complete linkage gives more compact clusters.

Exam Q&A:

Q5 — K-Means Feature Extraction (2 marks) **Q:** How can K-means be used as a feature extractor for a downstream classifier?

A: After fitting K-means with K clusters, represent each point x by: (1) its cluster membership (categorical feature), or (2) a vector of distances to all K centres ($\|x - \mu_1\|, \dots, \|x - \mu_K\|$). This creates a nonlinear feature space that can improve a downstream linear classifier.

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